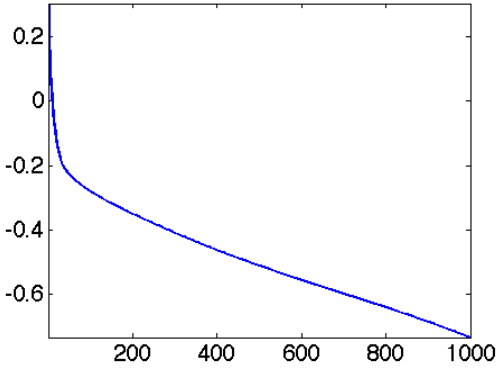
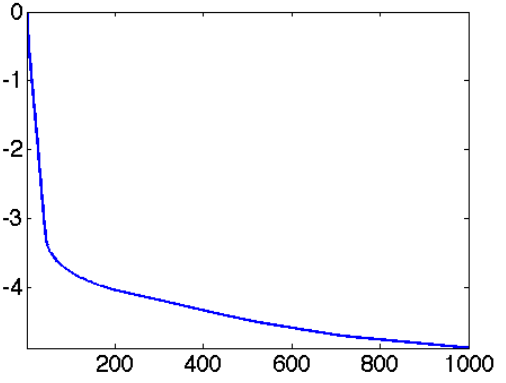


$\log_{10}(E(f^{(k)})/E(f^\star) - 1)$



$\log_{10}(\|f^{(k)} - f^\star\|/\|f_0\|)$